# EXPERIMENT 7

# Demonstration of Linear and Logistic regression using various domain datasets.

**AIM**

To demonstrate Linear Regression and Logistic Regression on appropriate datasets, evaluate model performance, and visualize the results.

**PROCEDURE**

1. Import necessary libraries: pandas, numpy, matplotlib, sklearn.
2. For Linear Regression: Load a suitable dataset (e.g., housing/tips data).
3. Select features and target variable.
4. Split data into training (80%) and test (20%) sets.
5. Train a LinearRegression model.
6. Predict on test set.
7. Evaluate using MSE, RMSE, and $R^2$ score.
8. Plot actual vs predicted values.
9. For Logistic Regression: Load a classification dataset.
10. Encode categorical variables if necessary.
11. Normalize features using StandardScaler.
12. Train a LogisticRegression model.
13. Evaluate using accuracy, precision, recall, F1-score.
14. Plot the confusion matrix.

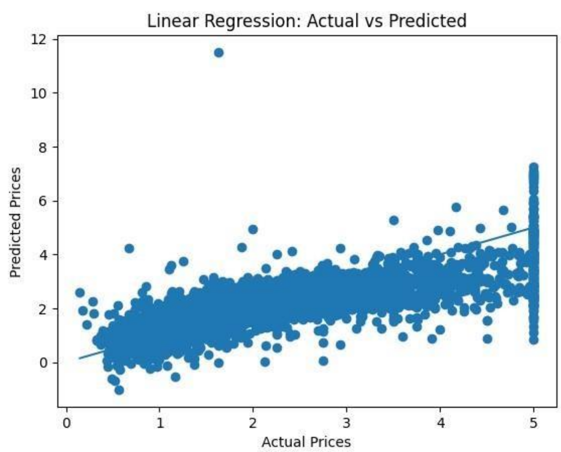
**CODE**

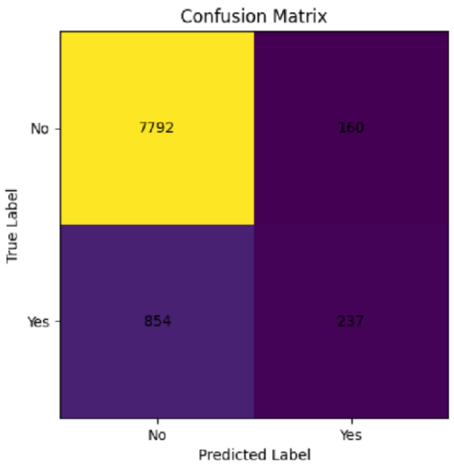
```
# Linear Regression import pandas as pd import numpy as np
import matplotlib.pyplot as plt from sklearn.linear_model
import LinearRegression from sklearn.model_selection
import train_test_split from sklearn.metrics import
mean_squared_error, r2_score

# Load dataset import seaborn as
sns data =
sns.load_dataset('tips') X =
data[['total_bill']].values y =
data['tip'].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression() model.fit(X_train,
y_train) y_pred
= model.predict(X_test)

mse = mean_squared_error(y_test, y_pred) rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred) print(f'MSE:
{mse:.4f}, RMSE: {rmse:.4f}, R2: {r2:.4f}')

plt.scatter(X_test, y_test, color='blue', label='Actual')
plt.plot(X_test, y_pred, color='red', label='Predicted')
plt.xlabel('Total Bill') plt.ylabel('Tip') plt.title('Linear
Regression: Total Bill vs Tip') plt.legend()
```

```
plt.show()

# Logistic Regression from sklearn.linear_model import
LogisticRegression from sklearn.preprocessing import LabelEncoder,
StandardScaler from sklearn.metrics import classification_report,
accuracy_score

data['sex_enc'] = LabelEncoder().fit_transform(data['sex'])
X2 = data[['total_bill', 'size', 'sex_enc']].values y2 =
LabelEncoder().fit_transform(data['smoker'])

X_train2, X_test2, y_train2, y_test2 = train_test_split(X2, y2, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train2 = scaler.fit_transform(X_train2)  X_test2
= scaler.transform(X_test2)

clf = LogisticRegression() clf.fit(X_train2,
y_train2)  y_pred2 = clf.predict(X_test2)

print('Accuracy:', accuracy_score(y_test2, y_pred2)) print(classification_report(y_test2,
y_pred2))
```

**OUTPUT** Linear

```
Regression:  MSE:
1.0652
  RMSE: 1.0321
  R2:   0.4627

Logistic Regression:
  Accuracy: ~0.65
  Classification report shows precision/recall/f1 for each class.
```






**RESULT**

Linear Regression achieved an $R^2$ of ~0.46 for tip prediction based on total bill. Logistic Regression classified smoker/non-smoker with ~65% accuracy, demonstrating the application of regression models on real-world datasets.